PSLN-344: portfolio review position

Every figure below is a group mean recomputed from the workbook unless another reduction is named. Survival percentages are the proportion of animals censored at the block's final observation day. Comparisons between arms are log-rank (Mantel-Cox), pairwise, unadjusted for multiplicity, which is the method the study states. Statements marked **[external]** are field knowledge, not findings in this package, and are flagged so the committee can tell the two apart.

Recommendation

Advance it.

In all six configurations where aged animals were vaccinated and then challenged with a virus of a different subtype, PSLN-344 separated from the vaccine alone (p = 0.0002 throughout) and from PSLN-212, which is the same liposome without the manganese (p = 0.0009 to 0.0151). On survival the aged animals were not distinguishable from young animals given PSLN-212 in any configuration (p = 0.060 to 0.317). PSLN-212 itself fails to separate from vaccine alone in three of the six (p = 0.088, 0.111, 0.066), so the manganese is carrying the effect rather than the liposome.

Two further blocks, S15-L and S15-N, are aged, vaccinated and challenged but state no challenge virus, so they cannot be classified as homo- or heterosubtypic and are not in the six. They point the same way: PSLN-344 6 of 7 in both, against 0 of 7 for sham in both and 0 of 7 and 2 of 7 for PSLN-212. Including them would not change the recommendation.

To be exact about what the six are, because the criterion matters more than the count: they are the four-arm cross-subtype challenge blocks on the S06 sheet. Two other blocks meet the subtype test and are still not among them. S15-H is aged, bivalent-vaccinated and challenged with rgH5N1, so its subtype is plainly derivable, but it carries no PSLN-212 arm and so cannot support the adjuvant-versus-parent comparison the six are counted for; it gives PSLN-344 6 of 7 against vaccine alone 0 of 7, which points the same way. S15-L and S15-N fail the subtype test itself. A reader who counts S15-H as a seventh favourable configuration is more complete than this document, not less, and should not be told otherwise.

The asset corrects an age-specific priming deficit disproportionately, but it is not exclusively an aged-animal effect, and the committee should hear it in that form rather than the stronger one. Three facts out of the same 2x2, on Ifnb1 at 16 hours (S01-A, group means of five animals, Welch t):

- The age deficit is real and large. Aged animals on PSLN-212 reach 16.64 against 37.60 in young, p < 0.0001.
- PSLN-344 restores aged animals to the young PSLN-212 level. 37.11 against 37.60, p = 0.857, which is as close to indistinguishable as anything in this package.
- But young animals given PSLN-344 go higher still, 50.70, significantly above both aged+PSLN-344 (p = 0.012) and young+PSLN-212 (p = 0.014).

So the adjuvant raises the interferon response in both cohorts. What is age-specific is the size of the increment, not its existence: across the five priming readouts it adds 1.65 to 24.35-fold inside aged animals against 1.02 to 1.35-fold inside young ones. That is still the profile we want for a 65-plus indication, and it is still the most defensible number set in the package, because it is the only comparison here that holds age and formulation apart. It will not support the claim that the asset does nothing in young animals, and a reviewer who checks the young+PSLN-344 arm will find that it does.

Two further qualifications. A sixth readout is also raised, the 72-hour interferon residual, and there the young gain is the larger of the two, 1.97 against 1.29. That one is a persistence signal rather than a benefit, so it does not help the case, but it is a second place where the age-specificity does not hold. Taken together, the age-specificity holds on the magnitude of the priming gain and nowhere else.

Two things the package does not support, stated up front because the committee will ask. It cannot tell us whether more antigen would do the same thing, and it cannot link any immune measurement to any animal's outcome.

The protection case

Nine challenge blocks sit on the S06 sheet. The set has to be derived before it can be counted.

S06-D is S06-C. All 28 time-and-event pairs are identical across all four arms. One experiment, entered twice, and counting both inflates the evidence. Nothing in the room says which label owns the data, and the body-weight blocks for the two challenge days are themselves distinct, so both experiments did run. The choice to retain the day-14 reading and set aside the day-3 one is therefore arbitrary; either could be the copy, and the day-3 rgH5N1 result should not be leaned on separately.

The two blocks at rows 69 to 70 are labelled S06-E and S06-F but hold the imprinting experiment, in 14-month-old animals given a bivalent vaccine containing both CA09 H1N1 and Aichi H3N2 and then challenged with PR8 H1N1 or Perth H3N2. Those challenges are within the subtypes the vaccine contains, so they belong to the imprinting question, not to the cross-subtype count.

That leaves six.

| Block | Vaccine | Challenge | Day | PSLN-344 | PSLN-212 | sham | untreated | |---|---|---|---|---|---|---|---| | S06-A | Aichi H3N2 | PR8 H1N1 | 14 | 85.7% | 14.3% | 0% | - | | S06-B | Aichi H3N2 | PR8 H1N1 | 3 | 71.4% | 0% | 0% | - | | S06-C | bivalent | rgH5N1 | 14 | 100% | 28.6% | 0% | - | | S06-E | Aichi H3N2 | PR8 H1N1 | 60 | 71.4% | 0% | 0% | 85.7% | | S06-F | bivalent | rgH5N1 | 60 | 71.4% | 14.3% | 0% | 100% | | S06-G | CA09 H1N1 | Perth H3N2 | 60 | 57.1% | 0% | 0% | 57.1% |

n = 7 per arm throughout. Reduction: proportion of rows censored at day 14.

| Comparison | S06-A | S06-B | S06-C | S06-E | S06-F | S06-G | |---|---|---|---|---|---|---| | PSLN-344 vs sham | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | | PSLN-344 vs PSLN-212 | 0.0083 | 0.0009 | 0.0068 | 0.0016 | 0.0094 | 0.0151 | | PSLN-344 vs young PSLN-212 | 0.317 | 0.141 | identical | 0.141 | 0.141 | 0.060 | | PSLN-212 vs sham | 0.0267 | 0.088 | 0.0024 | 0.111 | 0.066 | 0.0063 |

S06-G is not the weak end of a range. The three day-60 blocks each carry an arm of 20-month-old animals that received no vaccine and no virus. In S06-G that arm lost 3 of 7 by day 10, giving 57.1%, which is the same figure as the PSLN-344 arm (p = 0.855). The PSLN-344 arm therefore lost no more animals than an age-matched arm that met no virus at all. The workbook records only elapsed days and an event code, so it carries no cause of death and that is as far as the data goes. The same arm ran 6 of 7 in S06-E and 7 of 7 in S06-F, so S06-G drew a frailer cohort. Reporting 57.1% as a protection rate without that arm misstates the result.

Defence of the position. With n = 7 per arm, a non-significant comparison is not equivalence. Only S06-G is a tight match to its uninfected arm. In S06-E the PSLN-344 arm is one animal below it and in S06-F two below, and the test cannot resolve gaps that small in either direction. The honest claim is "not distinguishable at this sample size", and the next study is powered to do better.

Protection also appears at very short intervals. Challenge 3 days after a single dose gives 71.4% against 0% for sham (S06-B). In a separate experiment where challenge came 2 days after a single dose of rgH5N1 vaccine and animals were followed to day 21, PSLN-344 was 10 of 10 censored, PSLN-212 3 of 10 and sham 1 of 10 (S03-J, n = 10 per arm against a stated n of 9; one PSLN-344 row carries an event code with no day value and is counted as censored). At those intervals a vaccine-specific antibody response has not yet formed **[external: primary humoral responses require roughly one to two weeks]**, so whatever protects at day 2 or 3 is not antibody.

What is driving the protection

The chain is established by intervention, not by correlation. Each step was removed and the downstream readout fell.

Uptake into the alveolar compartment. Uptake of labelled liposome by alveolar macrophages was 43.10% in wild-type animals and 12.65% in Sftpa1-/-Sftpd-/- animals (S01-C, group means, n = 4). Carbenoxolone reduced recruited lung monocytes from 7.28% to 3.08% (S11-C, group means, n = 5); the room gives the agent's route and dose and not its mechanism, so read that step as gap-junction-dependent only on **[external: carbenoxolone is a gap-junction blocker]**.

Type I interferon. In the adoptive transfer, monocyte-derived dendritic cells were 19.70% from wild-type marrow and 2.09% from Ifnar-knockout marrow, and recombinant GM-CSF brought the knockout arm back to 12.40% (S02-H, group means, n = 5).

GM-CSF. Lung Gm-csf transcript at 36 hours was 8.74 for PSLN-344, 1.59 for PSLN-212 and 1.21 for sham, each relative to untreated animals (S02-F, group means, n = 5).

CCR2-dependent monocyte recruitment. INCB3344 reduced lung GZMB+ CD8 T cells from 6.77% to 1.39% (S02-I) and Ly6C+ macrophages from 15.35% to 2.52%, with monocytes from 9.86% to 3.26% (S07-B). Group means, n = 5.

Priming in the draining node. Pertussis toxin left proliferating transferred OT-I cells in lung at 0.93 of control (59.00% to 55.07%) while reducing them in the mediastinal node to 0.22 of control (30.90% to 6.83%) (S05-C, group means, n = 3).

Every one of those steps sits upstream of the adaptive response. The mechanism is improved priming, not an effector function of the adjuvant itself.

Where the chain stops. No blockade experiment carries a challenge arm. INCB3344 cuts the lung cytotoxic CD8 population 4.9-fold and Ly6C+ macrophages 6.1-fold, and no survival readout follows in either case, so the step from improved priming to improved survival is inferred rather than shown. That is the first thing the next study has to close.

An alternative the package already excludes. Protection at day 3 looks, on the cellular readouts alone, equally consistent with a non-specific antiviral state left by STING activation. Those readouts (NP-specific CD8 6.68% against 2.09%, S05-A, which is set aside below and is listed here only as part of the reading being tested; Ki-67+ CD8 9.38% against 3.48%, GZMB+ CD8 5.69% against 2.99%, all PSLN-344 against PSLN-212, group means) are real but sit inside the window where innate effects dominate **[external: primary CD8 responses to influenza in mice peak at days 7 to 10]**.

S15-H settles it. At the same 3-day interval it carries adjuvant alone, vaccine alone and the combination. Adjuvant alone kills every animal, 0 of 7 surviving with deaths on days 5, 5, 6, 6, 6, 7 and 8. Vaccine alone also loses all 7. The combination gives 6 of 7, with a single death on day 12. A non-specific antiviral state protects nobody here: antigen and adjuvant are each necessary and neither is sufficient. The day-3 protection is therefore adjuvant-dependent priming of an antigen-specific response, not a residual innate state, and the question does not need a new experiment. What still needs one is the step from that priming to survival in the mechanism-blockade arms, which is Study 1.

What the adjuvant actually adds

The package compares aged animals given PSLN-344 against young animals given PSLN-212. That comparison changes two things at once. In six labelled blocks, seven panels, the workbook also carries young animals given PSLN-344, which closes it.

| Readout | aged+PSLN-212 | aged+PSLN-344 | young+PSLN-212 | young+PSLN-344 | |---|---|---|---|---| | Ifnb1, 16 h (S01-A) | 16.64 | 37.11 | 37.60 | 50.70 | | Ifnb1, 72 h (S01-A) | 3.43 | 4.43 | 1.59 | 3.13 | | CD11b+ DC, d4 (S01-E) | 8.28 | 18.88 | 21.40 | 22.60 | | cDC1, 72 h (S01-F) | 22.80 | 37.52 | 43.42 | 44.26 | | cDC2, 72 h (S01-G) | 75.30 | 59.46 | 54.08 | 51.36 | | Antigen-bearing DC in node (S02-C) | 1.77 | 7.09 | 9.39 | 11.12 | | OT-I proliferation (S02-E) | 0.77 | 18.65 | 26.24 | 31.40 |

Group means, n = 5 throughout.

Adjuvant effect inside each cohort, PSLN-344 over PSLN-212. Six of the seven readouts go up in aged animals. Five are priming readouts: 2.23, 2.28, 1.65, 4.01 and 24.35-fold aged, against 1.35, 1.06, 1.02, 1.18 and 1.20-fold young. The sixth is the 72-hour interferon residual, 1.29-fold aged against 1.97-fold young, the one readout on which the young gain exceeds the aged one and the one where a larger number is not a better result. The cDC2 readout moves down in both, 0.79-fold aged and 0.95-fold young.

Two headline numbers change. Ifnb1 at 16 hours reads 0.99 of young on the diagonal, which looks like a match. Holding formulation constant it is 0.73. And the 72-hour residual reads 2.78 of young on the diagonal but 1.42 with formulation held, because young animals given PSLN-344 also fail to clear, 3.13 against 1.59 for young animals given PSLN-212.

The limit. Downstream of the dendritic cell compartment, no humoral or cellular endpoint carries a young-plus-PSLN-344 arm *alongside* the aged and young-PSLN-212 comparators, which is what a within-adjuvant contrast needs. The seven panels tabulated above are the whole of what does. S02-I does carry a young-plus-PSLN-344 arm, but only against its own CCR2-blocked control, so it supports no age contrast. For everything downstream of the dendritic cell compartment, age and formulation cannot be separated.

Adaptive output is real but below young. Across 18 endpoints with a young comparator and a known direction, every one points the same way. Sixteen where more is better fall below 1.0 (range 0.406 to 0.987) and two where more is worse sit above it. The roster is fixed as follows, so that every count and mean below can be rebuilt from the workbook:

| # | Endpoint | Block | Direction | ratio | |---|---|---|---|---| | 1 | Ifnb1, 16 h | S01-A | higher better | 0.987 | | 2 | Ifnb1, 72 h | S01-A | **higher worse** | 2.785 | | 3 | Ly6C+ monocytes, d4 | S01-D | higher better | 0.948 | | 4 | Alveolar macrophages, d6 | S11-A | higher better | 0.661 | | 5 | Serum IgG, H1N1 | S03-A | higher better | 0.618 | | 6 | Serum IgG, H3N2 | S03-E | higher better | 0.406 | | 7 | HAI, H1N1, geometric | S03-C | higher better | 0.758 | | 8 | IgG2a:IgG1, corrected | S03-B + S13-A/B | higher better | 0.721 | | 9 | Germinal centre B cells | S03-K | higher better | 0.666 | | 10 | ICOS-high Tfh | S04-A | higher better | 0.752 | | 11 | Tfr, d7 | S04-B | **higher worse** | 1.517 | | 12 | NP+ CD8, lung, d3 | S05-A | higher better | 0.678 | | 13 | NP+ CD8, lung, d7 | S05-A | higher better | 0.941 | | 14 | Ki-67+ CD8, d3 | S05-D | higher better | 0.944 | | 15 | GZMB+ CD8, d3 | S05-E | higher better | 0.592 | | 16 | IFN-gamma+ CD8, cross-subtype restim, d14 | S05-F | higher better | 0.693 | | 17 | IFN-gamma+ CD8, d60 | S05-G | higher better | 0.687 | | 18 | Lung resident memory CD8 | S05-I | higher better | 0.656 |

Means of the per-endpoint ratios: humoral 0.654 over rows 5 to 10, cellular 0.741 over rows 12 to 18. Three endpoints are effectively matched: monocyte recruitment at day 4 (0.95), day-7 NP-specific CD8 (0.94) and CD8 proliferation (0.94). Early interferon induction reads 0.99 on this comparator and would make a fourth, but that is the cross-adjuvant figure debunked above; held to one formulation it is 0.73 and does not belong in the matched set.

The roster matters because the counts depend on it. Tfr at day 30 (S04-C) is the same measurand as row 11, carries a young comparator, and sits at 1.893; MLN IgG in B220+ cells (S13-H) sits at 0.954, above all three matched endpoints. Neither is in the eighteen. Any count stated without the roster is unreconstructable and should not be relied on.

Against the age-matched comparator the same endpoints invert. H1N1 serum IgG is 18.06-fold over PSLN-212 (7891.6 against 437.0, S03-A, group means) and 0.62 of young. H1N1 HAI is 8.00-fold over PSLN-212 by geometric mean (367.58 against 45.95) and 0.76 of young. Which comparator we pick decides whether this asset looks transformative or disappointing, and for a 65-plus product the age-matched one is the relevant one.

Biomarkers, and what each could do in patients

Antigen-specific CD8 and tissue-resident memory. Carry forward as the efficacy marker. These are accepted correlates of cross-subtype protection **[external]**, so this package does not need to establish them. In the room they behave as the mechanism predicts: lung resident memory at 0.656 of young, interferon-gamma producing CD8 after cross-subtype restimulation at 0.693 at day 14 and 0.687 at day 60. The limit is access, not validity. The day-25 resident-memory readout is lung; the day-14 and day-60 restimulation readouts are mediastinal lymph node. Both compartments require tissue. In patients the role is mechanistic and pharmacodynamic, read in a controlled human infection study or from lavage in a dedicated cohort, not as a routine endpoint.

Antigen-specific antibody. Carry forward as immunogenicity and the regulatory bridge, not as the efficacy marker. HAI at the conventional threshold is an accepted correlate for matched strains **[external]**. It is peripherally accessible and routine, which is its value. But every antibody assay in this package was run against the antigens the vaccine contained, so none of them speaks to the cross-subtype claim, and no accepted antibody surrogate exists for cross-subtype protection **[external]**. Using titre as the efficacy readout would power the next study on the one thing the asset is not claiming.

Lung GM-CSF at 36 hours. Pharmacodynamic marker of adjuvant activity, preclinical only. It is antigen-independent, which makes it the cleanest readout of whether the adjuvant engaged, and it separates cleanly (8.74 against 1.59 and 1.21). It is also lung transcript, so there is no patient-accessible version of it. If we

want a clinical engagement marker we have to find one, and that is a gap rather than an asset.

Next studies

Study 1. Close the priming-to-protection step and settle the mechanism.

- *Establishes*: that the protection is monocyte and CD8-dependent, and that the chain reaches survival. The non-specific-antiviral-state alternative is already excluded by S15-H and does not need an arm here.
- *Endpoints*: survival to day 14 after challenge, primary. Lung antigen-specific CD8 and resident memory, secondary.
- *Population*: 20-month-old animals, intranasal, as in the mounted work.
- *Comparators*: PSLN-344-adjuvanted vaccine with CCR2 blockade, with CD8 depletion, and with vehicle; plus sham.
- *Readouts*: survival curves by log-rank, with the cellular readouts on the same animals so the immune measure and the outcome are linked per animal, which nothing in the current package does.
- *Timing*: challenge at day 3 and at day 14 in parallel, since the two intervals test different mechanisms. Cellular sampling at day 7 to 10, not day 3.

Study 2. Benchmark against the actual standard of care.

- *Establishes*: whether the adjuvant beats more antigen, which is the comparison the committee and any payer will make. High-dose inactivated vaccine is licensed for 65-plus **[external]**, and HA content is never varied anywhere in this package, so we currently cannot answer it.
- *Endpoints*: survival after cross-subtype challenge, primary. Antigen-specific CD8, resident memory and serum antibody, secondary.
- *Population*: 20-month-old animals.
- *Comparators*: standard-dose vaccine with PSLN-344; high-dose antigen without adjuvant, dose-ranged; standard-dose with PSLN-212; sham.
- *Readouts*: survival by log-rank; immune endpoints on the same animals.
- *Timing*: single dose, challenge at day 14 and day 60 to cover durability.

Study 3. Resolution kinetics, as the gating tolerability study.

- *Establishes*: when the innate response returns to baseline, which the package does not measure. Both S09-A blocks are labelled 16h in the workbook and the annotation repeats that label for both, so the timepoint of the second is an inference, not a stated fact: exponentiated, its Ifnb1 values reproduce the 72 h column of S01-A (4.01 against 4.43) while the first block reproduces the 16 h column (36.3 against 37.1). On that basis all seven analytes sit above the untreated reference (Ccl2 27.41, Ccl3 9.23, Tnf 8.44, Il6 7.19, Il10 4.16, Ifnb1 4.01, Ccl5 2.01, each 10 to the power of the mean log10 of three replicates), and no arm carries a later timepoint. Young animals given PSLN-344 also fail to clear, so this is a property of the formulation and not of the aged lung, which means it will not resolve itself in a younger population.
- *Endpoints*: time for Ifnb1, Tnf, Il10 and Ccl5 to return to the untreated reference. Lung histopathology and body weight, supporting.
- *Population*: 20-month-old animals, same strain and route.
- *Comparators*: age-matched PSLN-212 and age-matched untreated. The young arm is context, not the acceptance criterion.
- *Readouts*: the same seven-analyte panel reported on a linear scale with the reduction stated.
- *Timing*: sampling beyond 72 hours until the untreated reference is reached. The analyte panel fixes only 16 and 72 hours, so the design has to run past both. Other blocks do carry 36 h (S02-F, S02-C, S10-A), but none carries an analyte timepoint later than 72 h.

Considered and ruled out

S06-D, as a seventh configuration. Identical to S06-C in all 28 time-and-event pairs.

The two blocks at rows 69 to 70 of the S06 sheet, as cross-subtype evidence. Both challenge viruses fall inside the subtypes the bivalent vaccine contains. Retained for the imprinting question, where both PSLN-344 arms ran 85.7% whether or not the animals had prior PR8 exposure.

Antibody titre, as the readout to power the next study on. Largest effects in the package, but every antibody assay used the vaccine's own antigens.

The S03-B and S03-F subclass ratio panels, as printed. Recomputed per animal from the subclass data, the H1N1 ratio is 0.578 for PSLN-212, 2.144 for PSLN-344 and 2.972 for young, against 2.972, 0.578 and 2.144 as printed. The labels are shifted by one position. The reading is forced rather than preferred: under the printed assignment one arm's IgG2a plus IgG1 would have to exceed its own total IgG by 16.1-fold for H1N1 and 11.3-fold for H3N2, computing each per animal from the subclass titres of S13-A and S13-B over S03-A for H1N1, and S13-C and S13-D over S03-E for H3N2. Correctly paired the two subclasses account for 0.559, 0.346 and 0.551 of total IgG. The same shift appears in both panels.

The arithmetic mean for HAI. On S03-C it gives an PSLN-344 titre of 416.0 and a sham titre of 6.0, and the lowest dilution tested was 1 in 10, so a central estimate of 6.0 is below the floor of the assay. Reduced as a geometric mean throughout.

S05-A, as part of the day-3 mechanism case. The source gives two different immunisations for the same panel, ovalbumin in one statement and the bivalent influenza vaccine in the other, and nothing resolves it. The day-3 case rests on S05-D and S05-E, which are n = 5 and unambiguous. S05-I inherits the same ambiguity.

Extension beyond influenza. No non-influenza pathogen antigen anywhere. Ovalbumin is a model protein, not evidence of breadth.

Tolerability, checked rather than assumed. Both assays carry a positive control that moved. Body weight at day 10 (S08-D) was 99.78% of baseline for PSLN-344 against 90.65% for PSLN-212/R848, which is the arm that shows the assay can detect weight loss, and that group had 4 of 5 animals remaining at day 10 with a minimum of 84.7%. Central nervous system cytokines (S10-A) ran 1.13 to 1.59 for PSLN-344 against 2.68 to 5.50 for cholera toxin. Dosing sits at a plateau (S08-A): 20 micrograms gives 45.83 and 40 micrograms gives 50.23, a 9.6% increase for twice the dose. The open question is not acute tolerability but duration, which is Study 3.